

**BIRLA INSTITUTE OF TECHNOLOGY MESRA
OFF CAMPUS JAIPUR**



Project Report

Programme : **B. Tech.**
Semester : **VI**
Branch : **CSE**
Course Code : **EC 338**
Subject Name : **Embedded Systems Lab**

1. Introduction

In today's fast-paced world, monitoring the health of elderly people has become increasingly important, especially for those who live alone or without constant supervision. Regular checking of vital health parameters such as body temperature and heart rate plays a crucial role in early detection of potential health issues. However, due to age-related limitations or forgetfulness, elderly individuals may not consistently monitor their health.

This project presents an **IoT-based Health Monitoring System** designed using **Arduino, SIM800L GSM module, DHT11 sensor, and a pulse heart rate sensor**. The system provides a simple and efficient solution for monitoring basic health parameters and ensuring timely communication with caregivers.

The proposed system measures the **body temperature and pulse rate** of the user and sends this information via **SMS alerts** to a predefined contact using the GSM network. This enables remote monitoring of the elderly person's health condition without requiring internet connectivity.

Additionally, the system includes a **buzzer alert mechanism** that beeps at regular intervals, reminding the user to check their vital parameters. This feature ensures that the elderly person does not forget to take regular readings, promoting better health management.

Overall, this project aims to develop a **low-cost, reliable, and easy-to-use health monitoring system** that enhances the safety and well-being of elderly individuals while keeping their family members informed in real-time.

2. Problem Statement

Elderly people often require regular monitoring of vital health parameters such as **body temperature and heart rate** to detect potential health issues at an early stage. However, many elderly individuals live alone or without continuous supervision, making it difficult for caregivers or family members to track their health condition in real time.

Additionally, due to age-related challenges such as forgetfulness or lack of awareness, elderly individuals may not consistently check their health parameters at regular intervals. This can lead to delayed detection of abnormal conditions and increase the risk of serious health complications.

There is also a lack of simple, affordable, and easy-to-use systems that can both **remind elderly individuals to monitor their health** and **automatically communicate the collected data to a responsible person** without requiring complex technology or internet connectivity.

Therefore, there is a need for a **reliable, low-cost health monitoring system** that can:

- Remind the user to take regular measurements
- Measure key health parameters like temperature and pulse rate
- Send real-time alerts to caregivers through SMS

This project aims to address these challenges by developing an **Arduino-based health monitoring system integrated with GSM technology** for efficient and remote health tracking of elderly individuals.

3. Steps Towards Solution

To address the problem of irregular health monitoring and lack of real-time communication for elderly individuals, the following steps are implemented in this project:

1. Identification of Key Health Parameters

The first step is to identify essential health parameters that need regular monitoring. In this system, **body temperature and heart rate** are selected as they are basic indicators of a person's health condition.

2. Selection of Appropriate Sensors

Suitable sensors are chosen to measure the required parameters:

- **DHT11 sensor** for temperature measurement
- **Pulse sensor** for heart rate detection

These sensors are cost-effective and easy to interface with a microcontroller.

3. Microcontroller Integration

An **Arduino** is used as the central processing unit. It collects data from the sensors, processes it, and controls the overall functioning of the system.

4. Reminder Mechanism Implementation

A **buzzer** is integrated into the system to act as a reminder. It generates a beep sound at regular intervals to alert the elderly person to check their health parameters.

5. Data Processing and Validation

The Arduino reads and processes the sensor values to ensure accurate measurement of temperature and pulse rate before sending the data.

6. GSM Communication Setup

A **SIM800L GSM module** is connected to the Arduino to enable communication over the mobile network. It is programmed to send SMS alerts.

7. SMS Alert System

Whenever the user measures their health parameters, the system automatically sends an **SMS containing temperature and heart rate values** to a predefined phone number of a caregiver or family member.

8. Power Supply and System Testing

The system is powered using a suitable power source, and all components are tested to ensure proper working, accuracy, and reliability.

9. Final Integration and Deployment

All components are assembled into a complete system, making it compact and user-friendly for elderly individuals to use in daily life.

4. Circuit Diagram

The circuit diagram of the **IoT-based Health Monitoring System** consists of an Arduino connected to sensors, a GSM module, and a buzzer. Each component is interfaced in a specific way to ensure proper functioning of the system.

Components Connections

1. DHT11 Temperature Sensor

- VCC → 5V (Arduino)
- GND → GND (Arduino)
- Data Pin → Digital Pin (e.g., D2)

2. Pulse Heart Rate Sensor

- VCC → 5V (Arduino)
- GND → GND (Arduino)
- Signal Pin → Analog Pin (e.g., A0)

3. SIM800L GSM Module

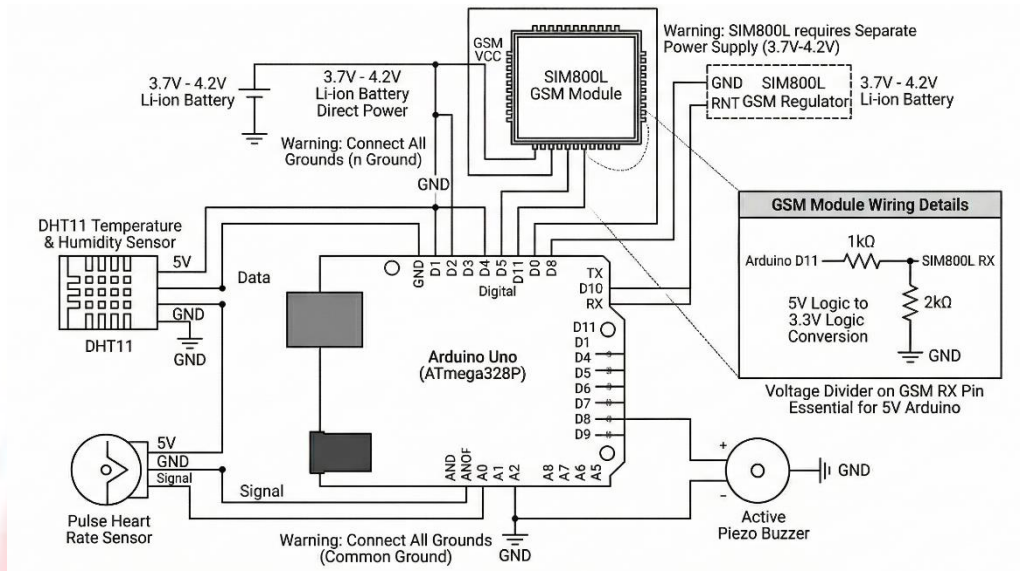
- VCC → External 3.7V–4.2V supply (Important: Do NOT connect directly to Arduino 5V)
- GND → GND (Arduino)
- TX → Arduino RX (e.g., D10)
- RX → Arduino TX (e.g., D11) (*use voltage divider for safety*)

4. Buzzer

- Positive Pin → Digital Pin (e.g., D8)
- Negative Pin → GND

5. Arduino (Controller)

- Acts as the central unit connecting all components
- Provides power and processes data



5. Conclusion

The **IoT-Based Health Monitoring System for Elderly** successfully demonstrates a simple, reliable, and cost-effective solution for monitoring basic health parameters such as **body temperature and heart rate**. By integrating sensors with an Arduino and a GSM module, the system is capable of collecting real-time data and sending it to a caregiver through SMS, ensuring continuous remote monitoring without the need for internet connectivity.

The inclusion of a **buzzer reminder system** further enhances the effectiveness of the project by prompting the elderly individual to regularly check their health parameters. This helps overcome the issue of forgetfulness and ensures timely monitoring.

Overall, the system improves the safety and well-being of elderly people, especially those living alone, by enabling early detection of potential health issues and maintaining communication with family members or caregivers.

Although the system is basic, it provides a strong foundation for future enhancements such as adding more sensors, integrating mobile applications, and enabling cloud-based data storage. Hence, this project proves to be a practical step toward developing advanced healthcare monitoring systems using IoT technology.